

MATH 203 HW 5

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October 2024

A

Question 3

Proof. Let $s_1 = \sqrt{2}$. For $n = 1, 2, 3, \dots$, define.

$$s_{n+1} = \sqrt{2 + \sqrt{s_n}}$$

It is clear that $\sqrt{2} < 2$. Therefore,

$$\sqrt{2 + \sqrt{s_n}} < \sqrt{2 + \sqrt{2}} < \sqrt{2 + 2} = 2$$

Hence, each $s_n < 2$. Also, each $s_n > 0$. So, $\{s_n\}$ is contained in the ball $B(0, 2)$. Consequently, $\{s_n\}$ is bounded. Also, $\{s_n\}$ is monotonically increasing, to show this we will proceed by induction. For $n = 1$, since $\sqrt{\sqrt{2}} > 0$,

$$\begin{aligned} s_1 &= \sqrt{2} \\ s_2 &= \sqrt{2 + \sqrt{\sqrt{2}}} > \sqrt{2} > s_1 \end{aligned}$$

Then assume $s_{n-1} < s_n$.

$$s_{n+1} = \sqrt{2 + \sqrt{s_n}} > \sqrt{2 + \sqrt{n-1}} = s_n$$

Hence, $\{s_n\}$ is monotonically increasing. By the monotone convergence theorem, $\{s_n\}$ converges. $\square$

Question 5

Proof. Let $\{a_n\}$ and $\{b_n\}$ be real sequences. Assume if $\limsup_{n \rightarrow \infty} a_n = \infty$, then $\limsup_{n \rightarrow \infty} b_n \neq -\infty$ and vice versa. Consider the case that one or both limsups

are ∞ .

$$\begin{aligned}\limsup_{n \rightarrow \infty} (a_n + b_n) &= \infty \\ \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n &= \infty\end{aligned}$$

The same holds true for $-\infty$. So, now we can examine the cases when both $\{a_n\}$ and $\{b_n\}$ have real limsups. Then, there exists a subsequence, $\{a_{n_k} + b_{n_k}\}$ that converges to $\limsup_{n \rightarrow \infty} (a_n + b_n)$. So,

$$\begin{aligned}\limsup_{n \rightarrow \infty} (a_n + b_n) &= \lim_{k \rightarrow \infty} (a_{n_k} + b_{n_k}) \\ \limsup_{n \rightarrow \infty} (a_n + b_n) &= \lim_{k \rightarrow \infty} a_{n_k} + \lim_{k \rightarrow \infty} b_{n_k} \\ \limsup_{n \rightarrow \infty} (a_n + b_n) &\leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n\end{aligned}$$

□

Question 16

(a)

Proof. Fix positive α , choose $x_1 > \sqrt{\alpha}$. Define $\{x_n\}$ as follows,

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right)$$

Take the case of $n = 1$,

$$\begin{aligned}x_2 &= \frac{x_1}{2} + \frac{\alpha}{2x_1} \\ x_2 &< \frac{x_1}{2} + \frac{\alpha}{2\sqrt{\alpha}} \\ x_2 &< \frac{x_1}{2} + \frac{\sqrt{\alpha}}{2} \\ x_2 &< \frac{x_1}{2} + \frac{x_1}{2} \\ x_2 &< x_1\end{aligned}$$

Also,

$$\begin{aligned}x_2 - \sqrt{\alpha} &= \frac{x_1}{2} + \frac{\alpha}{2x_1} - \sqrt{\alpha} \\ x_2 - \sqrt{\alpha} &> \frac{\sqrt{\alpha}}{2} + \frac{\alpha}{2x_1} - \sqrt{\alpha} \\ x_2 - \sqrt{\alpha} &> \frac{\sqrt{\alpha}}{2} + \frac{\alpha}{2x_1} > 0 \\ x_2 &> \sqrt{\alpha}\end{aligned}$$

Because $x_1 > 0$ and x_n is generated using only addition and multiplication with positive numbers, $x_n > 0$. Proceeding by induction, assume $x_n < x_{n-1}$. Then,

$$\begin{aligned} x_{n+1} - x_n &= \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \frac{1}{2} \left(x_{n-1} + \frac{\alpha}{x_{n-1}} \right) \\ x_{n+1} - x_n &= \frac{1}{2} \left((x_n - x_{n-1}) + \frac{\alpha(x_{n-1} - x_n)}{x_n x_{n+1}} \right) < 0 \\ x_{n+1} &< x_n \end{aligned}$$

Hence, $\{x_n\}$ is monotone decreasing. Additionally, assume $x_n > \sqrt{\alpha}$

$$\begin{aligned} x_{n+1} - \sqrt{\alpha} &= \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} \\ x_{n+1} - \sqrt{\alpha} &> \frac{1}{2} \left(\sqrt{\alpha} + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} \\ x_{n+1} - \sqrt{\alpha} &> \frac{1}{2} \left(\sqrt{\alpha} + \frac{\alpha}{x_n} \right) \\ x_{n+1} - \sqrt{\alpha} &> 0 \\ x_{n+1} &> \sqrt{\alpha} \end{aligned}$$

Thus, each $x_n > \sqrt{\alpha}$. So,

$$\begin{aligned} x_n \sqrt{\alpha} &> \alpha \\ \sqrt{\alpha} &> \frac{\alpha}{x_n} \end{aligned}$$

Also, $\{x_n\}$ is bounded as it is contained in the ball $B(0, 2)$ since $x_1 < 2$ and $\sqrt{\alpha} < x_n < x_1$. Hence, $\{x_n\}$ converges by the monotone convergence theorem. Thus, there exists some L such that,

$$\lim_{n \rightarrow \infty} x_n = L$$

x' Also, by theorem 3.3 and the recursion formula

$$\begin{aligned} \lim_{n \rightarrow \infty} x_n &= \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_{n-1} + \frac{\alpha}{x_{n-1}} \right) \\ L &= \frac{L}{2} + \frac{\alpha}{2L} \\ L^2 &= \alpha \\ L &= \sqrt{\alpha} \end{aligned}$$

□

(b)

Proof. Let $\varepsilon_n = x_n - \sqrt{\alpha}$. Then,

$$\begin{aligned}\varepsilon_{n+1} &= \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \\ \varepsilon_{n+1} &= \frac{1}{2x_n} (x_n^2 + \alpha) \\ \varepsilon_{n+1} &= \frac{\varepsilon_n^2}{2x_n} < \frac{\varepsilon_n^2}{2\sqrt{\alpha}}\end{aligned}$$

The last inequality follows as $x_n > \sqrt{\alpha}$

□

(c)

Proof. Let $\alpha = 3$ and $x_1 = 2$. Let $\beta = 2\sqrt{\alpha}$. Then,

$$\begin{aligned}\frac{\varepsilon_1}{\beta} &= \frac{2 - \sqrt{3}}{2\sqrt{3}} \\ \frac{\varepsilon_1}{\beta} &= \frac{2\sqrt{3} - 3}{6}\end{aligned}$$

So,

$$\begin{aligned}\frac{1}{10} - \frac{2\sqrt{3} - 3}{6} &= \frac{6 - 20\sqrt{3} + 30}{60} \\ \frac{1}{10} - \frac{2\sqrt{3} - 3}{6} &= \frac{9 - 5\sqrt{3}}{15} \\ \frac{1}{10} - \frac{2\sqrt{3} - 3}{6} &= \frac{3}{5} - \frac{\sqrt{3}}{3}\end{aligned}$$

$\frac{3}{5} > \frac{\sqrt{3}}{3}$ if and only if $(\frac{3}{5})^2 > (\frac{\sqrt{3}}{3})^2$ since both are positive.

$$\begin{aligned}\left(\frac{3}{5}\right)^2 &= \frac{9}{25} \\ \left(\frac{\sqrt{3}}{3}\right)^2 &= \frac{1}{3} \\ \frac{9}{25} &= \frac{27}{75} > \frac{25}{75} = \frac{1}{3}\end{aligned}$$

Hence, $\frac{\varepsilon_1}{\beta} < \frac{1}{10}$. Therefore,

$$\begin{aligned}\varepsilon_5 &< 2\sqrt{3} \left(\frac{\varepsilon_1}{2\sqrt{3}} \right)^{2^4} \\ \varepsilon_5 &< 2\sqrt{3} \left(\frac{1}{10} \right)^{16} \\ \varepsilon_5 &< \frac{2\sqrt{4}}{10^{16}} \\ \varepsilon_5 &< 4 * 10^{-16}\end{aligned}$$

Also,

$$\begin{aligned}\varepsilon_6 &< 2\sqrt{3} \left(\frac{1}{10} \right)^{2^5} \\ \varepsilon_6 &< \frac{4}{10^{32}} \\ \varepsilon_6 &< 4 * 10^{-32}\end{aligned}$$

□

Question 20

Proof. Let $\{p_n\}$ be a Cauchy sequence in the metric space X . Let $\varepsilon > 0$, consider the subsequence $\{p_{n_j}\}$ which converges to $p \in X$. There exists an $J \in \mathbb{N}$ such that if $j \geq J$ then,

$$d(p_{n_j}, p) < \varepsilon/2$$

there exists an $N \in \mathbb{N}$ such that if $m \geq n \geq N$, then,

$$d(p_m, p_n) < \varepsilon/2$$

In particular, there exists some j so that $n_j \geq N$ so that

$$d(p_{n_j}, p_n) < \varepsilon/2$$

Then,

$$d(p_{n_j}, p_n) + d(p_{n_j}, p) < 2(\varepsilon/2)$$

By the triangle inequality,

$$d(p_n, p) \leq d(p_n, p_{n_j}) + d(p_{n_j}, p)$$

Hence,

$$d(p_n, p) < \varepsilon$$

So, $\{p_n\}$ converges to p .

□

Question 21

Proof. Let X be a complete metric space. Let $\{E_n\}$ be a sequence of closed, nonempty, and bounded subsets of X such that $E_{n+1} \subset E_n$ and

$$\lim_{n \rightarrow \infty} \text{diam } E_n = 0$$

Take $a_n \in E_n$ and define $A_N = \{a_n \mid n \geq N\}$. Then

$$\text{diam } (E_n) \geq \text{diam } (A_n)$$

Hence, by the squeeze theorem,

$$\lim_{n \rightarrow \infty} \text{diam } (A_n) = 0$$

Hence, for each $\epsilon > 0$, there exists an $N \in \mathbb{N}$ so that if $m \geq n \geq N$, then,

$$d(a_n, a_m) \leq \text{diam } A_N < \epsilon$$

Then, A_n is a Cauchy sequence. So, because X is complete, A_n has a limit point. Since $A_n \subset E_n$ and E_n is closed, A_n is closed too. Hence, if x is the limit point of $\{A_n\}$ for all n , $x \in A_n$ for all n . Thus, $x \in \bigcap_{n=1}^{\infty} E_n$ so $\bigcap_{n=1}^{\infty} E_n$ is nonempty.

Now to show only one point exists. Proceeding by contradiction assume $a, b \in \bigcap_{n=1}^{\infty} E_n$ such that $a \neq b$. Then $a, b \in E_n$ for any n . Hence, by the limit above for all $\epsilon > 0$ and $n \in \mathbb{N}$,

$$\begin{aligned} d(a, b) &\leq \text{diam } E_n < \epsilon \\ d(a, b) &= 0 \\ a &= b \end{aligned}$$

Hence, a contradiction is achieved so only one point must exist in $\bigcap_{n=1}^{\infty} E_n$. $\square$

Question 22

Proof. Suppose X is a nonempty complete metric space. Let $\{G_n\}$ be a sequence of dense open subsets in X .

Let $F_n = G_n^c$. Let C be an arbitrary open set. Because each G_n is dense in X and C is open, C must contain at least a point from each G_n . Hence $C \not\subset F_n$ for any n .

So, there exists some $x \in C \setminus F_1$, which is an open set. Thus there exists

an $\varepsilon_1 > 0$ for which $B(x, \varepsilon_1) \subset C \setminus F_1$. Let $E_1 = B(x, \varepsilon_1/2)$.

Because $E_1 \subset C$, $E_1 \not\subset F_2$. So, there exists some $x_2 \in E_1 \setminus F_2$. Also, this set is also open. So, there exists an $\varepsilon_2 > 0$ so that

$$B(x_2, \varepsilon_2) \subset E_1 \setminus F_2 \subset C \setminus (F_1 \cup F_2)$$

Take $E_2 = B(x_2, \varepsilon_2/2)$. Continue to construct the sequence as follows for $n \in \mathbb{N}$.

Then, $\overline{E_1} \supset \overline{E_2} \supset \cdots \supset \overline{E_n}$ as

$$\overline{E_n} \supset E_n \supset B(x_{n+1}, \varepsilon_{n+1}) \supset \overline{B(x_{n+1}, \varepsilon_{n+1}/2)} = \overline{E_{n+1}}$$

The set of $\{\varepsilon_n\}$ is monotonically decreasing as each ball is contained within the previous. Hence, let $M > \varepsilon_1$ and then $B(x_1, M) \supset \overline{E_n}$ for any n . Hence, the set of $\{\overline{E_n}\}$ is bounded.

All that needs to be done before we can apply question 22 is to show that,

$$\lim_{n \rightarrow \infty} \text{diam } \overline{E_n} = 0$$

Let $\delta > 0$. Since the set of $\{\varepsilon_n\}$ is monotonically decreasing and in particular due to our choice of $\varepsilon/2$ for the balls, for each $n > N$,

$$\text{diam } \overline{E_{n+1}} < \frac{1}{2} \text{diam } \overline{E_n}$$

Then, there exists an N for which $\text{diam } E_N < \delta$. Consequently,

$$\lim_{n \rightarrow \infty} \text{diam } \overline{E_n} = 0$$

By question 22, $\bigcap_{n=1}^{\infty} \overline{E_n}$ contains a point. Hence, $\bigcap_{n=1}^{\infty} G_n$ contains at least a point and thus is nonempty. $\square$

Question 23

Proof. Let $\{p_n\}$ and $\{q_n\}$ be Cauchy sequences in a metric space X . Let $\varepsilon > 0$. Then, there exists $N_1 > 0$ such that if $m \geq n \geq N$ then,

$$d(p_n, p_m) < \varepsilon/2$$

Similarly, there exists $N_2 > 0$ such that if $m \geq n \geq N$ then,

$$d(q_n, q_m) < \varepsilon/2$$

Also, by the applying triangle inequality twice, for any m and n ,

$$\begin{aligned} d(p_n, q_n) &\leq d(p_n, q_m) + d(q_m, q_n) \\ d(p_n, q_n) &\leq d(p_n, p_m) + d(p_m, q_m) + d(q_m, q_n) \end{aligned}$$

Choose $N = \max\{N_1, N_2\}$. Then for $m \geq n \geq N$,

$$\begin{aligned} d(p_n, q_n) - d(p_m, q_m) &< \varepsilon/2 + \varepsilon/2 \\ d(p_n, q_n) - d(p_m, q_m) &< \varepsilon \end{aligned}$$

Hence, $\{d(p_n, q_n)\}$ is a Cauchy sequence. It is also a sequence of real numbers as distance is measured in real numbers. Hence, it converges as $\mathbb{R}$ is a complete metric space. $\square$

Question 24

(a)

Proof. Let $\{p_n\}, \{q_n\}$, and $\{a_n\}$ be Cauchy sequences. $\{p_n\}$ and $\{q_n\}$ are equivalent if,

$$\lim_{n \rightarrow \infty} d(p_n, q_n) = 0$$

To show that this is an equivalence relation we must show that it is reflexive, symmetric and transitive. First reflexivity.

Because $d(p_n, p_n) = 0$ by the properties of a metric, we can say

$$\lim_{n \rightarrow \infty} d(p_n, p_n) = \lim_{n \rightarrow \infty} 0 = 0$$

Thus, we say $\{p_n\}$ is equivalent to $\{p_n\}$ and thus the relation is reflexive.

Because $d(p_n, q_n) = d(q_n, p_n)$ by the properties of a metric, we can say,

$$\lim_{n \rightarrow \infty} d(p_n, q_n) = \lim_{n \rightarrow \infty} d(q_n, p_n)$$

Hence, the limit is symmetric.

Lastly, assume $\{p_n\}$ is equivalent to $\{q_n\}$ and $\{q_n\}$ is equivalent to $\{a_n\}$. Then,

$$\begin{aligned} \lim_{n \rightarrow \infty} d(p_n, q_n) &= 0 \\ \lim_{n \rightarrow \infty} d(q_n, a_n) &= 0 \end{aligned}$$

Let $\varepsilon > 0$. There exists N_1 so that if $n \geq N_1$

$$d(p_n, q_n) < \varepsilon/2$$

There exists N_2 so that if $n \geq N_2$

$$d(q_n, a_n) < \varepsilon/2$$

Take $N = \max\{N_1, N_2\}$ and let $n > N$. Then,

$$d(p_n, q_n) + d(q_n, a_n) < \varepsilon/2 + \varepsilon/2$$

By the triangle inequality,

$$d(p_n, a_n) \leq d(p_n, q_n) + d(q_n, a_n) < \varepsilon$$

Hence $\{p_n\}$ is equivalent to $\{a_n\}$ and transitivity holds. Thus the limit of the distance between the points being 0 defines an equivalence relation. $\square$

(b)

Proof. Let X^* be the set of all equivalence classes obtained in part (a). Let $P, Q \in X^*$ and let $\{p_n\} \in P$ and $\{q_n\} \in Q$. Define,

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(p_n, q_n)$$

This limit exists by question 23. Let $\{a_n\} \in P$ and $\{b_n\} \in Q$ so that $\{a_n\} \sim \{p_n\}$ and $\{b_n\} \sim \{q_n\}$. Hence,

$$\begin{aligned} \lim_{n \rightarrow \infty} d(p_n, a_n) &= 0 \\ \lim_{n \rightarrow \infty} d(q_n, b_n) &= 0 \end{aligned}$$

Let $\varepsilon > 0$. There exists N_1 so that if $n \geq N_1$

$$d(p_n, a_n) < \varepsilon/3$$

There also exists N_2 so that if $n \geq N_2$

$$d(q_n, b_n) < \varepsilon/3$$

Now examine $\lim_{n \rightarrow \infty} d(p_n, q_n)$. We know,

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(p_n, q_n)$$

Thus, there exists an $N_3 > 0$ such that if $n \geq N_3$ then,

$$d(p_n, q_n) - \Delta(P, Q) < \varepsilon/3$$

Let $N = \max\{N_1, N_2, N_3\}$. Then by the triangle inequality for $n \geq N$,

$$\begin{aligned} d(p_n, q_n) &\leq d(p_n, a_n) + d(a_n, b_n) + d(b_n, q_n) \\ d(p_n, q_n) - d(a_n, b_n) &< \varepsilon/3 + \varepsilon/3 \\ |d(p_n, q_n) - d(a_n, b_n)| &< 2\varepsilon/3 \end{aligned}$$

So using the reverse triangle inequality,

$$\begin{aligned}
d(p_n, q_n) - \Delta(P, Q) + |d(p_n, q_n) - d(a_n, b_n)| &< 2\varepsilon/3 + \varepsilon/3 \\
d(p_n, q_n) - \Delta(P, Q) + |d(a_n, b_n) - d(p_n, q_n)| &< \varepsilon \\
d(p_n, q_n) - \Delta(P, Q) + ||d(a_n, b_n)| - |d(p_n, q_n)|| &< \varepsilon \\
d(p_n, q_n) - \Delta(P, Q) + |d(a_n, b_n)| - |d(p_n, q_n)| &< \varepsilon \\
d(p_n, q_n) - \Delta(P, Q) + d(a_n, b_n) - d(p_n, q_n) &< \varepsilon \\
d(a_n, b_n) - \Delta(P, Q) &< \varepsilon
\end{aligned}$$

So,

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(a_n, b_n)$$

Now to show that this is a distance function. If $P \neq Q$ then

$$\Delta(P, Q) > 0$$

As $\lim_{n \rightarrow \infty} d(p_n, q_n) \neq 0$ and $d(p_n, q_n) \geq 0$. If $P = Q$ then,

$$\Delta(P, Q) = 0$$

As $d(p_n, p_n) = 0$. Next,

$$\Delta(P, Q) = \Delta(Q, P)$$

This is true because,

$$\lim_{n \rightarrow \infty} d(p_n, q_n) = \lim_{n \rightarrow \infty} d(p_n, q_n)$$

by the property of the distance function $d(x, y)$. Lastly, let $\{z_n\} \in Z \in X^*$. Then,

$$\Delta(P, Z) \leq \Delta(P, Q) + \Delta(Q, Z)$$

This is by the triangle inequality as,

$$\lim_{n \rightarrow \infty} d(p_n, q_n) \leq \lim_{n \rightarrow \infty} d(p_n, q_n) + \lim_{n \rightarrow \infty} d(q_n, z_n)$$

Thus, $\Delta(P, Q)$ is a distance function for X^* □

(c)

Proof. Take a Cauchy sequence $\{P_k\} \in X^*$. Choose Cauchy sequences $\{p_{k_m}\}$ in X so that $\{p_{k_m}\} \in P_n$. Let N_k be the first positive integer such that for all $k \in \mathbb{N}$, if $n \geq m \geq N_k$ then

$$d(p_{k_m}, p_{k_n}) < 2^{-k}$$

Let $p_k = p_{k_n}$ so that $d(p_k, p_{k_n}) < 2^{-k}$ for all $n > N_k$. Hence,

$$\lim_{k \rightarrow \infty} d(p_k, p_{k_n}) \leq 2^{-k}$$

For any l and k ,

$$d(p_k, p_l) \leq d(p_k, p_{k_n}) + d(p_{k_n}, p_{l_n}) + d(p_{l_n}, p_l)$$

□

(B)

Proof. Let X be a complete metric space with no isolated points. Suppose X is countable, number the elements of X as follows,

$$x_1, x_2, x_3, \dots, x_n, \dots \quad (1)$$

Let G_n be the complement of the n th point of X . Then, G_n is dense in X as the only point not in G_n is x_n and because there are no isolated points, every ball around x_n contains another point which must then be in G_n . Also, X is and only one point is removed, hence the ball around each point g of G_n can be of radius $d(g, x_n)$ and they will be fully contained in G_n . Thus G_n is open.

We now can apply Baire's Theorem from problem 22 to show that $\bigcap_{n=1}^{\infty} G_n$ is nonempty. This is a contradiction however as for any $x_n \in X$, $x_n \in G_n$ and as a consequence $x_n \notin G_n$. It follows that X must be uncountable. □

(C)

Proof. Let X be a complete metric space. Take $K \subset X$ to be a perfect subset. Then K is closed and for $k \in K$, $k \in K'$. Hence, for all $\varepsilon > 0$ and $k \in K$, the ball $B(k, \varepsilon)$ contains another point $p \in K$ so that $p \neq k$. This means that no point of K is isolated. K is closed subset of X and thus is a complete metric space in it's own right.

As such, we have shown that K is a complete metric space with no isolated points so we can apply part B. By part B, K is uncountable. □